

G-MAD: An Open-Source Toolkit for Synthetic RGB-T Military Asset Detection Data Generation

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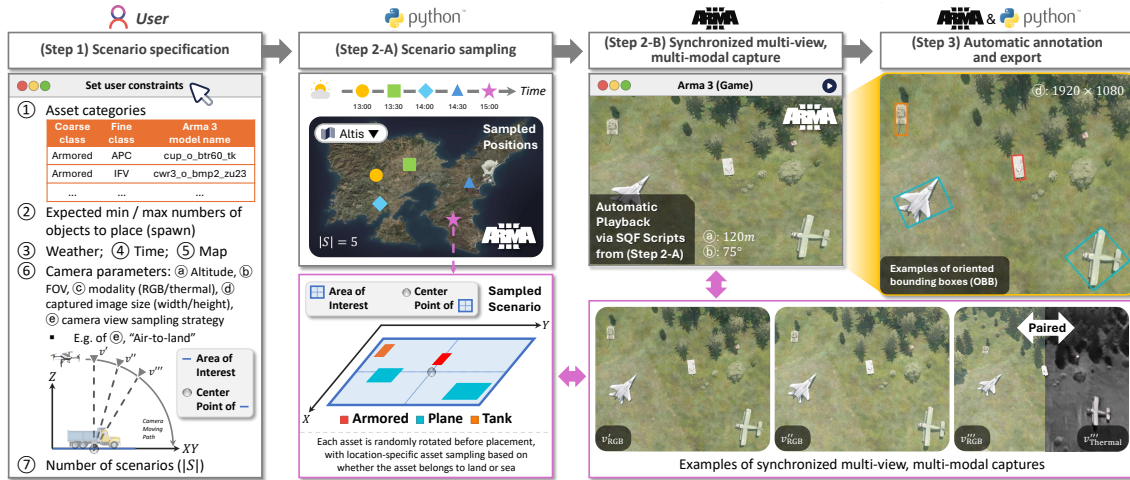


Figure 1: The overall architecture of G-MAD.

Abstract

Progress in military perception is often constrained not only by model design, but by who can access the data. Real military imagery is costly, sensitive, and operationally restricted, leaving many researchers without the means to study data-centric questions at scale, particularly in multi-modal settings such as RGB and thermal-infrared perception. We present G-MAD, an open-source software framework for generating controllable synthetic RGB-T military object detection datasets with automatic annotations using a game Arma 3. G-MAD enables users to flexibly compose military assets and environments, configure multi-view captures, and scalably generate datasets with diverse geometric, contextual, and cross-modal variation. By transforming restricted data collection problems into reproducible simulation workflows, the framework broadens participation in military vision research beyond highly specialized

organizations. We believe G-MAD provides a practical step toward democratizing synthetic-data-driven research for RGB-T military perception systems. The source code and further details are available at <https://unique-chan.github.io/G-MAD-Project>.

CCS Concepts

• Computing methodologies → Object detection.

Keywords

Synthetic Data Generation, Military Target, Object Detection, RGB, Thermal, Open-Source Toolkit, Deep Learning, Arma 3

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1 Introduction

Synthetic data has become a key component for scaling modern AI systems, mitigating challenges related to privacy, cost, and annotation [13, 22]. While recent approaches span a broad spectrum—from classical 3D graphics and simulation pipelines to neural rendering

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and generative models—not all methods provide the structured supervision and controllability required for perception tasks. In particular, simulation-based pipelines remain well suited for such settings, as they provide direct access to scene geometry, annotations, and sensor configurations [14].

In this context, commercial games have emerged as practical platforms for synthetic data generation [23]. While not originally designed for this purpose, they are widely adopted due to their rich pre-built assets, which eliminate the substantial cost of asset creation required in general-purpose simulation platforms such as Unity [6] and Unreal [7] Engine [8, 10, 30]. Among these, GTA V-based pipelines established a prominent line of work demonstrating large-scale automatic annotation for visual perception. Prior studies leveraged GTA V [5] and similar environments to construct datasets spanning multiple tasks like segmentation [23], crowd counting [25], anomaly detection [20], showing that game-based pipelines can support scalable and diverse supervision. More recent efforts further extend this paradigm to specialized tasks such as scene flow estimation [16] and HDR reconstruction [9]. In parallel, CARLA [14] has become a canonical programmable simulator for autonomous-driving research, supporting configurable sensors, environmental conditions, and standardized evaluation [15, 24, 29].

However, both game-based pipelines and programmable simulators are primarily designed for civilian domains such as urban driving, limiting their applicability to military perception scenarios involving specialized assets and sensing modalities. A game named Arma 3 [4] is an interesting alternative in this context because it already contains rich military assets and large terrains that are difficult to obtain in civilian simulators without heavy additional content creation [12, 18]. Note that anyone can obtain a permanent license of Arma 3 at a low cost. In this work, we present **G-MAD**, an efficient open-source toolkit that transforms Arma 3 into a structured dataset-generation framework for military object detection. Here, G-MAD stands for a **Data Generator for RGB-T Military Asset Detection**. It supports systematic control over military/civilian asset composition and environments for scalable data generation with diverse geometric contexts. We believe G-MAD facilitates controlled experimentation and lowers the barrier to entry for data-centric research in military target detection.

2 Architecture of G-MAD

G-MAD is designed as an end-to-end data generation pipeline that transforms a compact set of user-specified parameters into structured multi-modal (RGB-T) datasets. As illustrated in Fig. 1, the proposed framework decomposes the data generation process into three stages: (1) scenario specification, (2) scenario sampling and synchronized multi-view, multi-modal capture, and (3) automatic annotation and export. The overall pipeline is implemented in SQF, the native scripting language of Arma 3, allowing direct procedural control over scene composition, environmental configuration, in-game sensor control, and data export within the simulator.

In this work, we use **scenario** to denote a parameterized specification of data generation conditions, including object arrangement and environmental factors such as time, weather, and background. A scenario defines a fixed underlying configuration shared across multiple observations. Meanwhile, we use **scene** to denote a concrete capture instance generated from a scenario, corresponding to

a specific viewpoint and sensor configuration (e.g., camera pose and sensing modality such as RGB or thermal). Under this formulation, a single scenario produces multiple scenes by varying viewpoints and sensing modalities while keeping the object arrangement and environment unchanged, as shown in Fig. 1.

2.1 Scenario specification

G-MAD formulates data generation as a parameter-driven process. Rather than requiring users to manually construct individual scenarios, the proposed framework generates scenarios automatically from a compact set of high-level constraints provided by the user. These constraints include ① asset categories¹, ② the expected minimum and maximum numbers of objects to place, ③ weather, ④ time (day/night), ⑤ map, ⑥ camera sensor configuration (⑥-a. Altitude, ⑥-b. FOV, ⑥-c. RGB-thermal modality², ⑥-d. captured image size (width/height), ⑥-e. camera view sampling strategy per each scene), ⑦ number of scenarios. The scenario specification therefore defines only the admissible generation space rather than concrete object layouts or camera targets. Under these constraints, G-MAD automatically samples each scenario by randomly determining asset type (category) sampling, object arrangement, camera target point, and time, while keeping other factors such as weather and sensor settings fixed. For example, a user may specify that each generated scenario contains 5–10 non-overlapped assets within a predefined image resolution, under clear weather conditions, with a fixed camera altitude of 120 m and a field of view of 75°, using paired RGB-T captures from multiple viewing angles (20° and 40°).

2.2 Scenario sampling and multi-view, multi-modal capture

Given the user-specified constraints described in Sec. 2.1, G-MAD formulates data generation as a two-stage sampling process over scenarios and viewpoints. Let $\mathcal{S}$ denote the space of valid scenarios defined by user constraints, and $\mathcal{V}$ denote the space of possible viewpoints and sensing configurations. The data generation process can be expressed as $s \sim P(\mathcal{S} \mid \text{constraints})$, $v \sim P(\mathcal{V} \mid s)$, where each sampled pair (s, v) produces a single annotated scene.

This formulation decouples scenario-level variables (object arrangement, environment) from scene-level variables (camera viewpoint and sensing modality), enabling structured data generation. A single scenario can produce multiple consistent observations by varying viewpoints and sensing modalities while preserving the underlying layout. Specifically, for each sampled scenario, G-MAD generates multiple scenes by sampling camera poses around a target region and assigning sensing modalities such as RGB and thermal. This yields a structured dataset of size $|\mathcal{S}| \times |\mathcal{V}|$, where scenes derived from the same scenario share identical object configurations but differ in viewpoint and modality.

¹To enable *coarse-to-fine classification*, the framework provides support for hierarchical class labeling. (E.g., Helicopter - Observing Helicopter) We additionally provide a *pre-defined table that assigns each of the 641 unique Arma 3 asset models* to one of the 12 coarse classes described in Sec. 4, enabling users to immediately generate data; kindly refer to the “classes/CLASSES.csv” of our code. By referring to this table, users can create data for their desired assets. Moreover, indeed, users are *not limited to military assets*; they can also import civilian equipment into Arma 3. For further details, please refer to the “README.md” in our code and the official Arma 3 website.

²Basically, *white-hot thermal images* are generated using the Arma 3 TI engine. If specific spectrum (e.g., MWIR) is required, you may use image-to-image translation techniques (e.g., [12]), to convert the generated images into the target spectrum.

Table 1: Comparison of Arma 3-based synthetic data generation frameworks.

	[1]	[2]	G-MAD (Proposed)
Scene configuration and sampling			
Multi-map support	×	×	○
Max objects per image	≤ 1	≤ 1	≥ 1
Region of interest selection	Manual	Manual	Automatic (Constraint-aware)
Camera viewpoint sampling	Predefined orbit	Unconstrained random	User-defined sampling model
Weather variation	Fog, Sun, Rain	Fog, Sun, Rain	Fog, Overcast, Sun, Rain, Snow
Occlusion check between camera and object	×	○	○
Annotation and output			
Annotation method	Post-hoc image differencing	Post-hoc image differencing	Engine-native ground truth
Bounding box support	×	HBB	HBB, OBB
Resolution (Image size)	Fixed	Fixed	Configurable
Resolution scaling (supersampling)	×	×	○
Execution and automation			
MATLAB dependency	○	×	×
Execution scheme	GUI-dependent control	GUI-dependent control	GUI-independent launch
Temporary file cleanup	Manual	Manual	Automatic
Workstation availability during generation	Unavailable	Unavailable	Available

Camera viewpoints are defined through a user-configurable sampling model, allowing *controlled multi-view data generation*. The current G-MAD framework offers two default motion modes, “air-to-air” and “air-to-land” for users who prefer not to design a custom sampling model. In “air-to-air”, the camera moves laterally while maintaining a relatively stable altitude, producing sweeping aerial observations; see Fig. 2. In “air-to-land”, both altitude and lateral position vary following a parabolic trajectory, resulting in oblique views directed toward the ground; see Fig. 1.

Importantly, the viewpoint sampler is designed to be modular and extensible³. Rather than relying on a fixed trajectory, G-MAD defines camera poses through a parameterized sampling routine, allowing users to incorporate custom motion patterns with minimal modification, e.g., circular orbits or task-specific reconnaissance paths. Such flexibility enables users to tailor viewpoint distributions to specific downstream tasks.

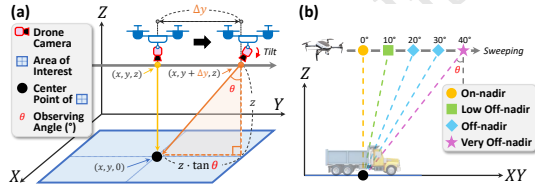


Figure 2: Illustration of one of the default camera viewpoint sampling methods in G-MAD: “air-to-air”. (a) Geometric parameterization of camera motion around the area of interest, defined by the observing angle and lateral displacement. (b) Viewpoint sweep from nadir to off-nadir.

This structured generation process provides two key benefits for learning. First, it enables consistent supervision across viewpoints, allowing models to learn viewpoint-invariant representations from multiple observations of the same underlying scene. Second, it naturally supports aligned multi-modal data generation, where RGB and thermal images share identical annotations, facilitating supervised cross-modal and fusion-based learning.

³Users can implement their own camera viewpoint sampling model in “src/my_scene_generator/util.py”. Specifically, please carefully update “get_all_cam_and_target_points()” and “get_all_cam_and_target_points_diff()”. For more details, please see “README.md” of our code.

2.3 Automatic annotation for object detection

G-MAD implements a direct annotation pipeline by querying object-level geometric information from the Arma 3 engine and projecting it into the image plane. Unlike prior approaches that rely on foreground-background image differencing, our framework retrieves precise 3D bounding box information from the Arma 3’s engine and transforms it into 2D annotations aligned with each captured scene. This ensures high-quality labels without manual intervention.

Formally, let an object in a scenario be associated with a set of 3D bounding box vertices. These vertices are projected onto the image plane using the camera parameters corresponding to each sampled viewpoint. The resulting 2D coordinates are used to generate both horizontal and oriented bounding boxes (HBB/OBB), providing richer supervision for detection tasks.

Furthermore, G-MAD incorporates automatic filtering of invalid annotations, such as objects fully occluded by other scene elements, as determined by the absence of a direct line-of-sight between the camera and the object surface. This reduces label noise of ghost bounding boxes compared to pipelines such as [1] that do not explicitly handle occlusion.

2.4 Comparison with existing Arma3-based open source data generation frameworks

To clarify the contribution of G-MAD, we compare it with two representative open-source Arma 3-based data generation frameworks in Tab. 1. Prior approaches [1, 2] are limited in both data diversity and systematic generation: they assume at most one object per image, rely on manual or inflexible camera setup, and provide little support for research-oriented, controlled dataset construction. Another key limitation is execution control. Existing methods depend on GUI-level automation via click-based actions to launch and operate Arma 3, making the pipeline brittle to UI state changes and user interference. In addition, their annotations are obtained through image differencing⁴, which is cumbersome and error-prone for multi-objects. Furthermore, this inherently limits its extensibility to data generation frameworks for video-level tasks, such as tracking.

In contrast, G-MAD enables constraint-aware multi-object generation, GUI-independent execution, and Arma 3 engine-native

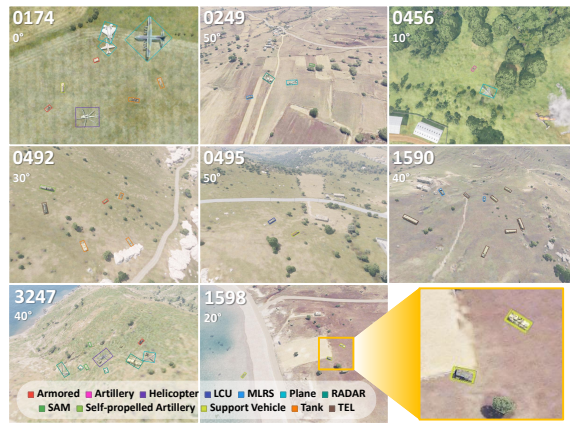
⁴In [1, 2], the object localization is inferred indirectly from image differencing between two distinct renders, one with the object present and the other with the object absent.

Table 2: Cross-angular evaluation results (AP_{50:95}) on AMOD.

Train	Test						
	0°	10°	20°	30°	40°	50°	Mean
0°	51.16	49.62	47.44	43.07	36.88	26.60	42.46
10°	50.92	50.01	48.04	44.79	38.76	28.39	43.49
20°	50.51	49.88	48.57	45.64	40.13	32.71	44.57
30°	48.67	48.27	47.42	45.69	41.46	33.66	44.20
40°	46.56	46.67	46.41	44.80	42.24	37.07	43.96
50°	38.23	38.20	39.04	39.29	37.95	36.49	38.20
All	56.18 (+5.02)	54.75 (+4.74)	53.93 (+5.36)	51.03 (+5.34)	47.67 (+5.43)	42.20 (+5.13)	50.96 (+6.39)

Table 3: AMOD vs. ImageNet pretraining on two real-world benchmarks, DIOR-R (Visible) and HIT-UAV (Thermal).

Pretrained from	Finetuned for DIOR-R					Finetuned for HIT-UAV			
	Airplane	Ship	Vehicle	Windmill	Mean	Person	Car	Others	Mean
ImageNet	71.88	81.02	42.59	57.07	63.14	85.30	81.29	57.12	74.57
AMOD	79.71	89.04	53.21	57.36	69.83 (+6.69)	87.68	82.15	61.40	77.08 (+2.51)

**Figure 3: RGB Examples from the AMOD dataset.****Figure 4: Qualitative analysis of an AMOD-trained detector on unseen real-world RGB imagery containing military equipment on real-world RGB satellite photographs released through public news coverage of the Russia-Ukraine war.**

annotation of both HBBs and OBBs. Unlike [1, 2], our work further supports multi-map generation, configurable resolution with supersampling, and automatic cache cleanup, establishing Arma 3 as a structured and scalable data generation framework.

3 Application of G-MAD

To demonstrate the research utility of G-MAD, we highlight three representative application settings enabled by our framework: 1) multi-view learning, 2) cross-modal RGB-T learning and fusion, and 3) pretraining for real-world object detection. Because G-MAD generates multiple synchronized observations from a shared underlying scenario while allowing controlled variation of viewpoint, modality, and environmental factors, it provides a practical foundation for studying data-centric learning problems.

4 Benchmarking test

To validate the effectiveness of G-MAD, we conduct a set of benchmarking experiments focusing on two aspects, 1) and 3), aligned with Sec. 3. We construct a dataset using G-MAD, referred to as Aerial Military Object Detection (AMOD)⁵, which consists of synchronized RGB-T multi-view aerial observations with OBB annotations of following categories: *Armored, Artillery, Helicopter, LCU, MLRS, Plane, RADAR, SAM, Self-propelled Artillery, Support Vehicle, Tank*, and *TEL*; see Fig. 3. As a baseline detector, we adopt Oriented R-CNN [28] with a Swin-S [21] backbone, following prior work [17, 19, 26, 31] on aerial object detection, using MMRotate [11, 32].

Multi-view learning. We first evaluate the effect of multi-view supervision using synchronized observations generated from the same scenarios. As shown in Tab. 2, while single-view models degrade under unseen viewing angles, multi-view training significantly improves generalization, achieving a mean AP₇₅ of 52.29 compared to 41.75 for the best single-view model (+10.54). This suggests that viewpoint diversity itself can be a major contributor to object detection.

Pretraining for real-world detection. We further evaluate a synthetic-to-real transfer setting, where models are pretrained on RGB and thermal variants of AMOD and fine-tuned on real-world RGB and thermal benchmarks, respectively. As shown in Tab. 3, AMOD pretraining consistently outperforms ImageNet initialization, improving mean AP₅₀ from 63.14 to 69.83 on DIOR-R (+6.69) and from 74.57 to 77.08 on HIT-UAV (+2.51). This suggests that AMOD provides transferable pretraining signals for both visible and thermal real-world aerial detection⁶.

Unseen real-world inference of military targets. Since authoritative real-world datasets for military target detection are difficult to obtain, we further perform an unseen real-world inference test from public news coverage, strictly without any fine-tuning. As shown in Fig. 4, the AMOD-trained detector⁷ still identifies plausible military targets on previously unseen real images.

5 Conclusions

In this work, we presented G-MAD, a scalable and configurable framework for generating synthetic RGB-T military asset detection data with synchronized multi-view capture and automatic annotation. Using G-MAD, we also constructed and released AMOD, an aerial military object detection dataset, and preliminary experiments with AMOD suggest that the proposed framework supports both controlled benchmarking and transfer to real-world settings. As future work, we plan to extend G-MAD into a more advanced data generation platform for UAV+UGV detection scenarios. We also aim to broaden the framework beyond detection toward more challenging tasks, including segmentation and tracking. *G-MAD is intended for non-commercial research use only. Further information regarding the licensing terms is provided on our official website.*

⁵The AMOD benchmark generated using G-MAD is described in more detail in our benchmark manuscript; see [3] of our official website. Tabs. 2 and 3, and Figs. 3 and 4 are reproduced from [3]. Note that this manuscript focuses on the software architecture, whereas [3] focuses on the benchmark itself.

⁶For pretraining, we perform a one-time style transfer of our AMOD toward the target RGB/thermal domain, respectively; the detailed pipeline is in [3] and our website.

⁷During AMOD training, only the RGB split is leveraged, incorporating DOTA [27]-style transfer; kindly refer to [3] for details.

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